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Government of India
Ministry of New and Renewable Energy

Atal Akshay Urja Bhawan
Lodhi Road, New Delhi 110003
Dated: 24th December 2025

OFFICE MEMORANDUM

Subject: Guidelines on Inverter Communication Devices, Dongles, Dataloggers, and Remote Monitoring Systems (RMS) for Communication with the Centralised IoT SCADA Platform under PM Surya Ghar: Muft Bijli Yojana - reg.

This refers Ministry's OM of even number dated 21.7.2025 prescribing compliance requirements for inverters and communication devices used under PM Surya Ghar: Muft Bijli Yojana (PMSG: MBY). It was also mentioned in the OM that Ministry is in the process of finalising the detailed vendor neutral and open communication protocol based data communication and security guidelines for RTS monitoring and control through National portal.

2. Subsequently, draft guidelines on RMS for Inverter Communication Devices, Dongles, Dataloggers with centralised platform were issued for stakeholders' consultation vide Ministry's OM of even number dated 21.7.2025.
3. Based on the comments received from various stakeholders, the Ministry has finalized the Guidelines on Inverter Communication Devices, Dongles, Dataloggers, and Remote Monitoring Systems (RMS) for Communication with the centralised IoT SCADA Platform under PM Surya Ghar: Muft Bijli Yojana.
4. The above mentioned Guidelines are attached for information and appropriate action by all concerned.

Vikram 24.12.25
(Vikram Dhaka)
Scientist -D

Encl: As above

To,
All concerned.

Copy to CMD, REC Limited, the NPIA for PMSG: MBY for information and appropriate action.



सत्यमेव जयते

नवीन एवं
नवीकरणीय ऊर्जा मंत्रालय
MINISTRY OF
**NEW AND
RENEWABLE ENERGY**

GUIDELINES ON INVERTER COMMUNICATION DEVICES, DONGLES, DATALOGGERS, AND REMOTE MONITORING SYSTEMS

**For Communication with the Centralised IoT SCADA Platform
under PM Surya Ghar: Muft Bijli Yojana**

24 December 2025

Ministry of New & Renewable Energy (MNRE)
Government of India

Document Overview

These guidelines outline the communication and security architecture required for seamless integration of Remote Monitoring Systems (RMS) of grid-connected rooftop solar (RTS) power plants, with the Centralised IoT SCADA Platform. It serves as a comprehensive reference for developers and system integrators to ensure standardized, secure, and interoperable data exchange across the solar energy ecosystem.

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1. BACKGROUND

The Ministry of New and Renewable Energy (MNRE) is implementing the PM Surya Ghar: Muft Bijli Yojana (PMSG: MBY) targeting installation of RTS plants for one crore households across the country with a cumulative capacity of 30 GW. Simultaneously, the Ministry has been taking strides to strengthen the self-reliance in supply chain for RTS components, and ensure transparency, and effective monitoring under the scheme.

The integration of around one crore RTS plants under PMSG: MBY introduces significant grid stability and cybersecurity challenges. Inverter communication modules that transmit data to servers outside India not only pose risks of unauthorized control but also threaten national energy sovereignty by exposing sensitive consumption and generation data. Millions of Inverters communicating with third party OEM servers can be exploited to manipulate inverter operations, potentially leading to coordinated disruptions in power generation and grid instability.

In this regard, multiple rounds of deliberation have been undertaken with major stakeholders to implement a framework to track generation of RTS installations as well as for the centralization of RTS data on a National Software Platform hosted on servers located in India and managed by MNRE or any other Government Agency to ensure monitoring and control of millions of inverters aggregating GW of capacity remains with Government agencies considering Grid Security and Stability.

In order to address these concerns, MNRE issued an **Office Memorandum dated 21st July 2025 on “Compliance requirements for inverters and communication devices used under PM Surya Ghar: Muft Bijli Yojana”**, mandating the use of M2M (Machine to Machine) SIM communication protocol for all Inverter Communication devices/dongles/dataloggers deployed under the scheme.

Subsequently, MNRE released **Draft Guidelines on RMS for inverter communication devices/dongles/data-loggers with the Centralised Platform**, inviting comments and suggestions from stakeholders. These draft guidelines detailed a comprehensive communication and security architecture covering RMS registration, MQTT-based messaging, device authentication, encryption, and standardised data exchange formats.

Accordingly, based on (i) Insights gained from **pilot implementations**, (ii) Feedback received during **stakeholder consultations**, including inverter manufacturers, RMS vendors, system integrators, and technology providers, and (iii) Evolving requirements related to **grid operations, cybersecurity, and interoperability**, MNRE has finalised **GUIDELINES ON INVERTER COMMUNICATION DEVICES, DONGLES, DATALOGGERS, AND**

REMOTE MONITORING SYSTEMS for communication with the Centralised IoT SCADA Platform.

These Guidelines aim to establish a **vendor neutral, uniform, secure, interoperable, and scalable national framework** for real-time monitoring and data aggregation of RTS plants, while ensuring **data sovereignty, vendor neutrality, and long-term extensibility** in alignment with India's renewable energy and digital public infrastructure objectives.

2. OBJECTIVES OF THE GUIDELINES

The objectives of these Guidelines are to:

1. **Establish Open, standardized, and vendor-neutral communication Connectivity and Protocols** for Remote Monitoring Systems (RMS) / Inverter communication devices / dongles / dataloggers deployed of grid-connected RTS installations
2. **Enable seamless, reliable, and scalable integration** of field-deployed solar assets with the **Centralised IoT SCADA Platform** through uniform protocols, message structures, and data models.
3. **Ensure data security, integrity, and confidentiality** through strong encryption, certificate-based authentication, controlled authorization, and transaction-level security mechanisms.
4. **Facilitate real-time and near-real-time monitoring** and diagnostics, to support operational efficiency, grid stability, and system reliability.
5. **Ensure data sovereignty and regulatory oversight** by mandating centralized data exchange through Government-authorized infrastructure hosted within India.
6. **Provide a future-ready framework** that can support additional devices, advanced analytics, control functions, and evolving regulatory or grid requirements without fundamental architectural changes.

3. DEFINITIONS AND ABBREVIATIONS

3.1 Definitions

- **Centralised IoT SCADA Platform:**
Centrally deployed and managed SCADA Software platform for monitoring, configuration and operations of inverters via RMS/Datalogger deployed in RTS installations.
- **RMS/Datalogger:**
A field-installed device responsible for data acquisition, data aggregation from field equipment, local storage, protocol translation, encryption, and secure transmission of data to the Centralised IoT SCADA Platform.
- **Field Devices:**
Equipment installed at the RTS plant level, including inverters, energy meters, string combiner boxes, and sensors etc.

- **MQTT (IEC 20922):**
A lightweight publish-subscribe messaging protocol used for communication between RMS and the Centralised IoT SCADA Platform.

3.2 Abbreviations

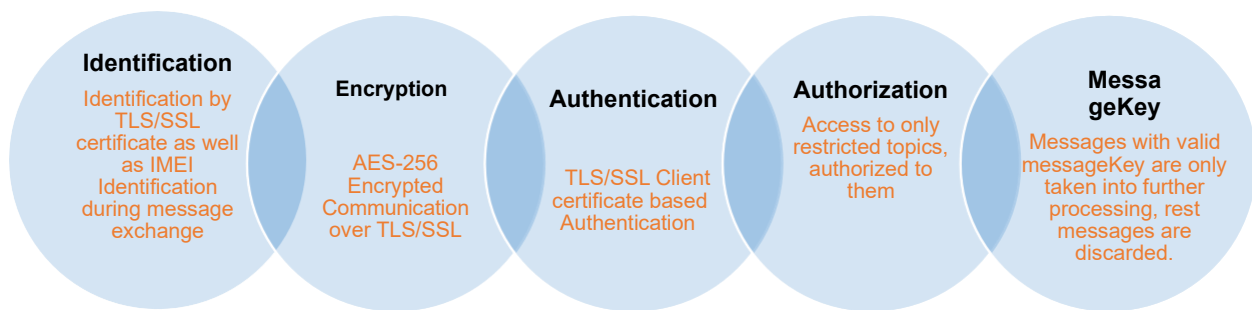
Abbreviation	Description
APN	Access Point Name
DAQ	Data Acquisition System
DLMS	Device Language Message Specification
IG	Grid-Tied Inverter
IH	Hybrid Inverter
IMEI	International Mobile Equipment Identity
IoT	Internet of Things
IS	String Inverter
JSON	JavaScript Object Notation
LAN	Local Area Network
MITM	Man-in-the-Middle
MNRE	Ministry of New & Renewable Energy
MQTT	Message Queuing Telemetry Transport
MSGID	Message Transaction Identifier
OEM	Original Equipment Manufacturer
RMS	Remote Monitoring System
RTC	Real-Time Clock
RTS	Rooftop Solar
SCADA	Supervisory Control and Data Acquisition
SCB	String Combiner Box
TLS/SSL	Transport Layer Security / Secure Sockets Layer
VD	Virtual Device
VPN	Virtual Private Network

4. SECURITY ARCHITECTURE

This section highlights the communication security architecture between RMS and Centralised IoT Platform. With this security and architecture, third parties are unable to intercept or “sniff” the encrypted data. This prevents ISPs, employers, local network administrators and cybercriminals from being able to perform “packet sniffing” to access what the traffic contains. It also protects against Man-in-the-Middle (MITM) attacks. This implements Private TLS/SSL VPN to ensure the highest level of security.

In addition to this, use of “MessageKey” in every message exchange shall help restrict spammers and Bots. Such a “MessageKey” based mechanism will provide transaction level security, which is required for remote operations and configurations.

In the event of any need to change the configuration of a RMS, the Server will publish a MessageKey. This will be received by the RMS/datalogger as it would have subscribed to the “MessageKey” topic of respective IMEI number e.g. “rtsg-1/ongridrooftop/{imei}/messagekey/sub”



To avoid MITM attacks, RMS/datalogger shall execute the commands only when either previous or current MessageKey of command & RMS/datalogger message key matches, else to be discarded.

5. RMS REGISTRATION

This section details how individual RMS shall be registered and communicate securely with the Centralised IoT Platform.

- a. Unique credentials would be generated against each vendor (RTS Installer) on Centralised IoT Platform, underneath that account multiple credentials can be generated.
- b. Log into Centralised IoT Platform from “Vendor” organization user and Register a new IMEI Number
- c. A unique Certificate, Client ID, Username and Password is generated based on “IMEI Number, Solution, Vendor ID”
- d. Configure the credentials and parameter register address against JSON keywords into RMS/Datalogger using local configuration web page hosted on built-in webserver of RMS/datalogger
- e. Detailed Message JSON keywords are explained from Annexure-1 to Annexure-12
- f. Detailed procedure of Registration and Issuance of certificate for RMS/Datalogger is explained in the Annexure-13

- g. Detailed procedure of certificate Renewal/Expiry is explained in Annexure – 14

6. MQTT TOPIC STRUCTURE

This section defines the different topic structure for communication between RMS and Centralised IoT Platform through Device Management Module.

RMS will publish and subscribe to their respective topics only; authorization of topic shall be done against unique credentials.

Application Version	Solution	IMEI	Message Type	Publish/Subscribe
rtsg-1	Ongridrooftop	{IMEI}	Info	sub
	Solarkw		messagekey	sub
			Heartbeat	pub
			Data	pub
			Ondemand	sub
			Config	sub

Sample Topic structure for on grid Rooftop shall be:

rtsg-1/Ongridrooftop/{IMEI}/info/sub

Multiple sub-topics will be formed for communication between RMS and Centralised IoT Platform

1. **Info:** Default Topic To exchange RMS configuration details
2. **MessageKey:** To exchange MessageKey at every interval of 15/30/60 minutes
3. **Heartbeat:** To update RMS health indicators at frequent, configurable intervals.
4. **Data:** To exchange data related to RMS Monitoring parameters in “**push mode**”
 - Push data Periodically
 - Push data on Event/Notification
 - History Missing Data Push Mode: History data will be identified against “**index**”
5. **Ondemand:** To exchange data between RMS and Server in “**Command on Demand**” Mode
 - Each “On Demand” message will have two transactions: Commands, Response.
 - On demand command and response will be tracked against a common “**MSGID**”.
 - On demand message can be used to read and write with two command types
 - Command: “**Read**” - Replace each key in the JSON received from the server with the corresponding value from the RMS, and send the updated JSON back to the server.
 - Note: Handshake parameters, including msgid and others, must be returned to the server exactly as received, without modification.
6. **Config:** To update configurable parameters of Device, which is similar to Ondemand but will be used only for configurable parameters of Device, this implements “**Configuration over the air**”
 - Command: “**Read**” - Replace each key in the JSON received from the server with the

- corresponding value from the RMS, and send the updated JSON back to the server.
- Command: **“Write”** - After executing the command based on the key–value pairs received in the JSON, send the updated JSON back to the server to confirm successful execution.
- Note: Handshake parameters, including msgid and others, must be returned to the server exactly as received, without modification.

7. COMMUNICATION MODES

- **Push on Periodic Interval:** In this mode deployed RMS shall transmit data of Multiple devices and sensors on different configurable time intervals such as Inverter at every 5 minutes, Energy Meter data (Optional) at every 15 minutes, String Combiner Box data (Optional) at every 10 minutes
- **Push on Event:** RMS shall detect various configurable alarm or event conditions such as Inverter On/Off Status, Fault or Trip status etc. and it shall transmit data immediately to the server
- **On Demand Read:** In this mode, User will send command to RMS to get data as and when required and RMS will send the required data to server immediately
- **Configuration read/write:** Using this mode, user will be able to read and change configurable parameters remotely such as updating periodic interval, alarm limits, server parameters etc.

8. COMMUNICATION CONNECTIVITY & PROTOCOLS

- Inverter Communication: RMS to Inverter communication shall be established using MODBUS RTU protocol in case of RS485 serial communication and Modbus TCP Protocol in case of Ethernet/WiFi LAN Communication supported by all leading Inverter Manufacturers.
- Energy Meter Communication (Optional): RMS to Energy Meter communication such as Bidirectional (Revenue) Meter, Solar Generation (Audit) Meter shall be established using IS15959/IEC62056 i.e. RS-232 DLMS protocol supported by all leading Meter Manufacturers.
- Local Connectivity for Configuration of devices/dongles/dataloggers/RMS: In case of replacement of SIM/RMS device/Inverter/Energy Meter, it shall be possible to configure and map new component using web page hosted on built-in webserver of RMS/datalogger via LAN (Ethernet/WiFi) connectivity.
- RMS to Server Communication – IEC20922 (MQTT): RMS to Server Communication shall be established using IEC20922 (MQTT) protocol which is well accepted IoT protocol across the globe and supported by all leading IT as well as OT companies for Smart Grid, Smart RE Applications etc.
- In case of unavailability of cellular network, devices/ dongles/ dataloggers/ RMS should store RTC timestamped data locally and on availability of network it should push data to central server. Local data storage should be possible for at least 60 days in case of unavailability of cellular network.
- Above mentioned communication connectivity and protocols required as a part of

RMS/Datalogger for the integration of various RTS components is explained in Fig.1

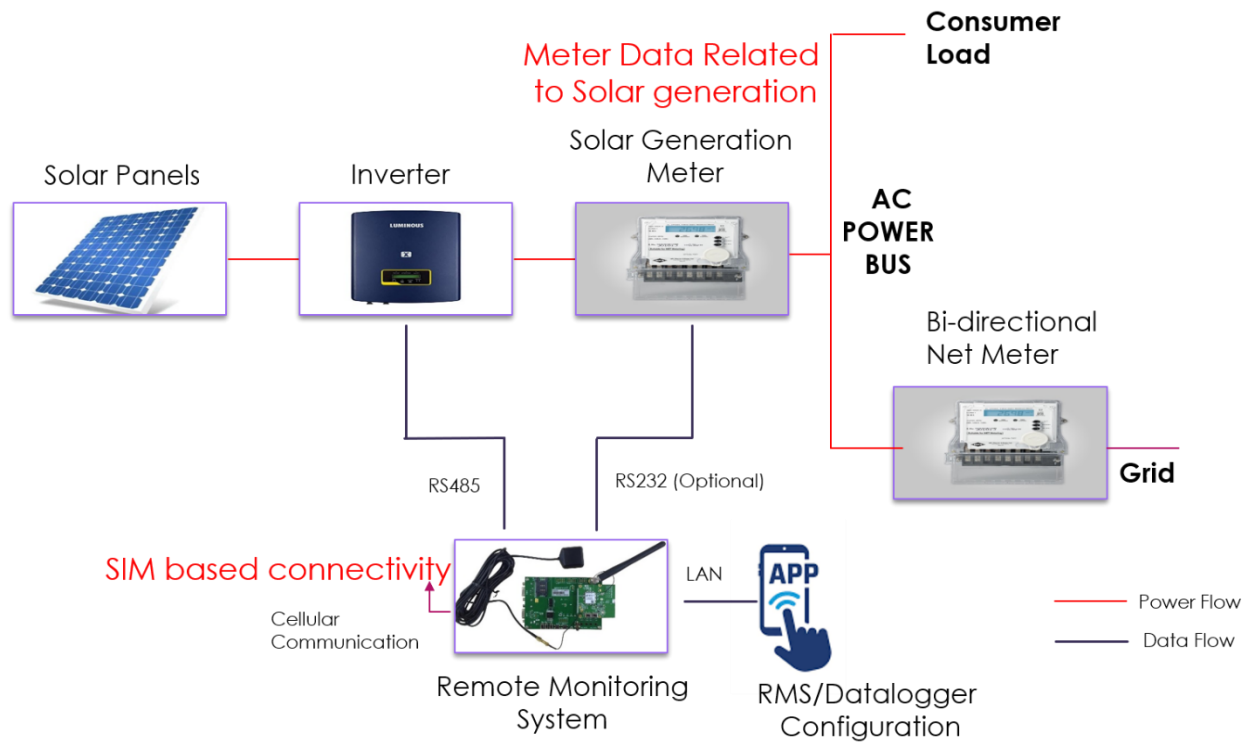


Fig-1 —: Communication Connectivity and Protocols of RMS for RTS Plant

9. JSON MESSAGE KEYWORD FORMATION

RTS plant comprises of many communicable equipment such as Inverter, Energy Meter, String combiner box etc. This equipment's are installed and measured at plant level as well Block level, accordingly all the parameters to be measured must be uniquely identified as per Fig.2 keyword designer to be referred for JSON annexure preparation.

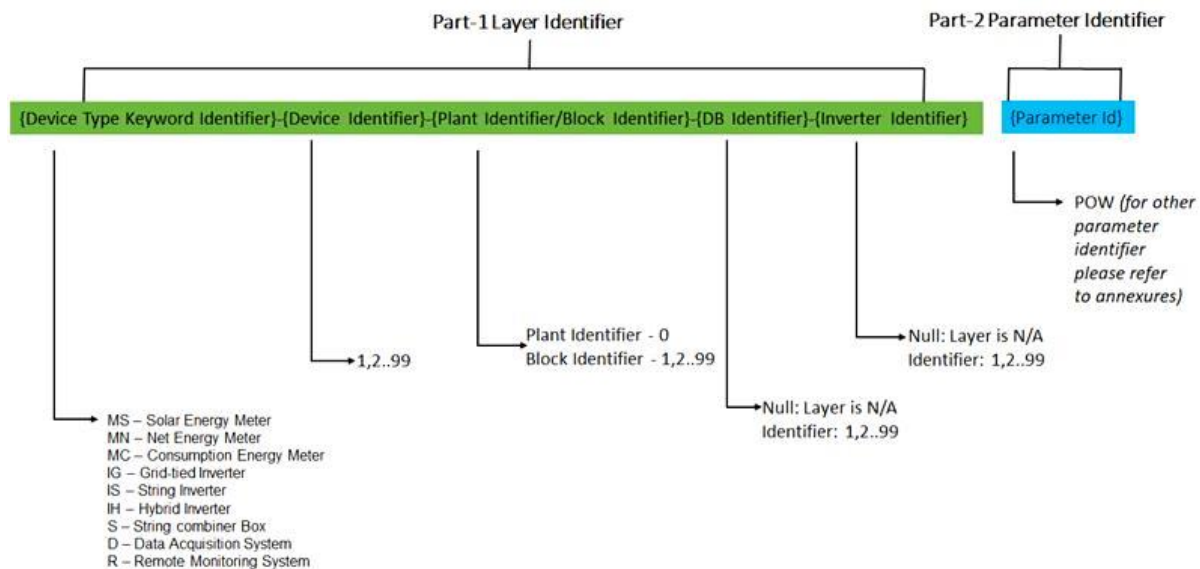


Fig-2 –: JSON Message Keyword Formation

Part-1 Layer Identifier: {Device Type Keyword Identifier}-{DeviceId}-{Plant/BlockId}- {DB Id}- {Inverter Id}-

Part-2 Parameter Identifier: {Parameter Id} .

Sample Parameter Description	Sample Unique JSON Keyword
Total Active Power of 1 st String Inverter of Plant	IS-1-0- - -POW
Total Active Power of 2 nd Grid tied Inverter of 3 rd DB of 2 nd Block (With DB)	IG-2-2-3- -POW <i>As this parameter belong to inverter so no need of duplicating inverter identifier. So Null value between two hyphens to be mentioned.</i>
Total Active Power of 3 rd Grid tied Inverter of 2 nd Block (Without DB)	IG-3-2- - -POW <i>As DB layer is not mentioned, DB id is not included so Null value between two hyphens to be mentioned. As this parameter belong to inverter so no need of duplicating inverter identifier. So, Null value between three hyphens in this required parameter</i>
Total Active Power of 95 th Hybrid Inverter of the plant	IH-95-0- - -POW
Total Active Power of 7 th Net Energy meter of entire plant	MN-7-0- - -POW
Total Active Power of 3 rd Solar Energy meter of 1 st Block	MS-3-1- - -POW
Total Active Power of 9 th Consumption Energy Meter of 1 st block of 2 nd DB	MC-9-1-2- -POW
Total Active Power of 5 th String combiner box of 2 nd inverter of 1 st DB of 2 nd Block (With DB)	S-5-2-1-2-POW
Total Active Power of 5 th String combiner box of 2 nd inverter of 2 nd Block (Without DB)	S-5-2- -2-POW <i>As DB layer is not mentioned, DB id is not included so Null value between two hyphens to be mentioned.</i>

Note: For parameter id refer to Annexures of different device types

10. MQTT MESSAGE STRUCTURE

This section details message structure exchanged between RMS and Centralised IoT Platform through Device Management Module. This message structure comprises of System reserved keywords (common for all equipment JSON formats - header) and equipment wise parameters.

Sample Keyword	Sample Description	Sample Value											
IMEI	Unique Identification of RMS – required to ensure registered source of data	863287049443888	System Reserved Header keywords, which are to be used as is in JSON exchange between RMS and Centralised IoT Platform, without prefix of Part-1 Layer Identifier										
VD	Virtual device/group – required for grouping of parameters based on update interval / subsystems such as String inverter, String Combiner Box, Energy Meter (Optional), Data Acquisition System etc.	2											
	<table><tr><td>Example of Virtual Device Number (say)</td><td>Device Type</td></tr><tr><td>0</td><td>RMS Health</td></tr><tr><td>2-4</td><td>Energy Meters (Optional)</td></tr><tr><td>5-24</td><td>Inverter, String Inverter / String Combiner Box (Optional)</td></tr><tr><td>25-27</td><td>Data Acquisition System</td></tr></table>			Example of Virtual Device Number (say)	Device Type	0	RMS Health	2-4	Energy Meters (Optional)	5-24	Inverter, String Inverter / String Combiner Box (Optional)	25-27	Data Acquisition System
	Example of Virtual Device Number (say)			Device Type									
	0			RMS Health									
	2-4			Energy Meters (Optional)									
	5-24			Inverter, String Inverter / String Combiner Box (Optional)									
25-27	Data Acquisition System												
MSGID	MSGID (Message Transaction ID) is required for 'Ondemand' or 'Config' message types, as well as for requests, responses, acknowledgments, and feedback. MSGID need to be compulsorily send with index value for backfill data. For normal data, the MSGID should be left blank.	123456789											
COMMAND	Read/Write - Applicable only in case of “Ondemand”/ “Config” message Type	Read											
TIMESTAMP	RTC timestamp of RMS against all parameters of VD / group (YYYY-MM-DD HH:MM: SS)	“2025-12-02 12:15:08”											
STINTERVAL	Periodic interval at which RMS shall store and transmit data to the server. (in minutes)	15											
DATE	local storage date – required as a reference to fetch data from local storage (YYMMDD)	251202											
INDEX	Local storage Index – required as a reference to fetch data from local storage	5											
MAXINDEX	Local storage maximum index of local storage date – required to calculate missing index	96											
LOAD	Local storage retrieval command & status	0											
PMKEY	Previous Message Key	123456											
CMKEY	Current Message Key	123456											
Parameter-1 Parameter-2 Parameter-3 Parameter-1 Parameter-n	Equipment-wise Keywords for multiple Parameters, These Keywords are to be prepared using: Part-1 Layer Identifier: {Device Type Keyword Identifier}-{DeviceId}-{Plant/BlockId}-{DB Id}- {Inverter Id}- Part-2 Parameter Identifier: {Parameter Id} - As per Annexure.												

11. ANNEXURES

11.1 Annexure-1: Single Phase-String Inverter (IS)

Annexure-1: Single Phase-String Inverter (IS)					
Message Name : Periodic Push Inverter Data					
Message Format : JSON					
Message Type : Data					
Message Command Flow : Not Applicable for Data periodic Push					
Message response Flow : RMS -> Centralised IoT Platform					
Message Medium : Cellular					
Keyword structure :“{Part-1: Layer Identifier}- {Part-2: Parameter Identifier}”					
Combined Sample Keyword : IS-1-2-3- -POW					
Sample Keyword Part-2: Parameter Identifier	Sample Description		Sample Unit	Sample Value	Sample Requirement in JSON
VD	Virtual Device Index/Group		-	5	Must Have
TIMESTAMP	RTC timestamp of RMS against all parameters of VD/group YYYY-MM-DD HH:MM: SS		-	2025-04-22 17:58:25	Must Have
MAXINDEX	maximum index of local storage date		-	96	Must Have
INDEX	reference of local storage		-	7	Must Have
LOAD	Local storage retrieval command & status		-	0	Must Have
STINTERVAL	Periodic interval at which RMS shall store and transmit data to the server. (in minutes)		-	15	Must Have
MSGID	Message Transaction Id - required for “Ondemand”/” Config” message type, request/response/acknowledgement/ Feedback		-	-	Must Have
DATE	local storage date YYMMDD		-	250422	Must Have
IMEI	IMEI No. of First Sim to be considered always for unique identity of RMS		-	123456123456123	Must Have
ASN_31	Inverter Serial Number		-	31123450	Must Have
	RMS	0			
	DAQ	1-9			
	Meter (Optional)	21-29			
	Inverter, String Inverter /String Combiner Box (Optional)	31-50			
IST	Inverter Status OFF: 0, ON: 1, FAULT :2, Other: 3		-	1	Must Have
DCV1	DC-1 Voltage (Similar for Up to N DC Channel Voltage)		Volts	233.10	Must Have

DCI1	DC-1 Current (Similar for Up to N DC Channel Current)	Amp	4.02	Must Have
DCKW1	DC-1 Power (Similar for Up to N DC Channel Power)	kW	0.93	Must Have
VN	Phase to Neutral voltage	Volts	248.80	Must Have
I	Phase current	Amp	3.50	Must Have
POW	Active Power	kW	0.88	Must Have
TKWH	Today Generated Energy	kWh	8.30	Must Have
TON	Today on Time of Inverter To convert Today running minutes into hours, simply divide by 60 $250/60 = 4.16$ Hrs.	Hrs.	7.45	Must Have
LKWH	Lifetime Generated Energy	kWh	9257.90	Must Have
LON	Lifetime running hours To convert lifetime running minutes into hours, simply divide by 60 $250/60 = 4.16$ Hrs.	Hrs.	16861.03	Must Have
TEMP	Inverter Temperature	Deg C	54.43	Must Have
FREQ	Frequency	Hz	49.99	Must Have
PF	Power Factor	-	0.91	Must Have
APOW	Apparent power	kVA	0.82	Must Have
RPOW	Reactive power	kVAr	0.05	Must Have
FT1	Fault-1 Normal: 0 VAC High: 1 VAC Low: 2 Grid Voltage Fail: 3	-	1	Must Have
FT2	Fault-2 Normal :0 Frequency High: 1 Frequency Low: 2	-	2	Must Have
FT3	Fault-3 Normal :0 AC Output overload :1 PV Short Circuit :2 AC Short circuit fault: 3 Leakage Current High: 4	-	0	Must Have
FT4	Fault-4 Normal :0 High Temperature: 1	-	0	Must Have
FT5	Fault-5 Normal: 0 Other Fault: 1	-	1	Must Have

11.2 Annexure-2 Three Phase - String Inverter (IS) / Three Phase-Grid Tied Inverter (IG)

Annexure-2 Three Phase - String Inverter (IS) / Three Phase-Grid Tied Inverter (IG)					
Message Name : Periodic Push Inverter Data					
Message Format : JSON					
Message Type : Data					
Message Command Flow : Not Applicable for Data periodic Push					
Message response Flow : RMS -> Centralised IoT Platform					
Message Medium : Cellular					
Keyword structure :“{Part-1: Layer Identifier}- {Part-2: Parameter Identifier}”					
Combined Sample Keyword : IS-1-2-3- -POW (for String Inverter) IG-1-2-3- -POW (for Central Inverter)					
Sample Keyword Part-2: Parameter Identifier	Sample Description	Sample Unit	Sample Value	Sample Requirement in JSON	
VD	Virtual Device Index/Group	-	5	Must Have	
TIMESTAMP	RTC timestamp of RMS against all parameters of VD/group YYYY-MM-DD HH:MM: SS	-	2025-04-22 17:58:25	Must Have	
MAXINDEX	maximum index of local storage date	-	96	Must Have	
INDEX	reference of local storage	-	7	Must Have	
LOAD	Local storage retrieval command & status	-	0	Must Have	
STINTERVAL	Periodic interval at which RMS shall store and transmit data to the server. (in minutes)	-	15	Must Have	
MSGID	Message Transaction Id - required for “Ondemand”/” Config” message type, request/response/acknowledgement/ feedback	-	-	Must Have	
DATE	local storage date YYMMDD	-	250422	Must Have	
IMEI	IMEI No. of First Sim to be considered always for unique identity of RMS	-	123456123 456123	Must Have	
ASN_32	Inverter Serial Number		-	34123450	Must Have
	RMS	0			
	DAQ	1-9			
	Meter (Optional)	21-29			
	Inverter, String Inverter /String Combiner Box (Optional)	31-50			
IST	Inverter Status OFF: 0, ON: 1, FAULT :2, Other: 3	-	1	Must Have	

DCV1	DC-1 Voltage (Similar for Up to N DC Channel Voltage)	Volts	342.38	Must Have
DCI1	DC-1 Current (Similar for Up to N DC Channel Current)	Amp	6.01	Must Have
DCKW1	DC-1 Power (Similar for Up to N DC Channel Power)	kW	2.07	Must Have
RPHV	R phase voltage	Volts	254.10	Must Have
RPHI	R phase current	Amp	2.77	Must Have
POWR	R phase Active Power	kW	0.70	Must Have
YPHV	Y phase voltage	Volts	252.10	Must Have
YPHI	Y phase current	Amp	2.77	Must Have
POWY	Y phase Active Power	kW	0.69	Must Have
BPHV	B phase voltage	Volts	253.90	Must Have
BPHI	B phase current	Amp	2.77	Must Have
POWB	B phase Active Power	kW	0.71	Must Have
POW	Total Active Power	kW	1.95	Must Have
TKWH	Today Generated Energy	kWh	17.40	Must Have
TON	Today ON Time of Inverter To convert Today running minutes into hours, simply divide by 60 $250/60 = 4.16$ Hrs.	Hrs.	8.16	Must Have
LKWH	Lifetime Generated Energy	kWh	50628.45	Must Have
LON	Lifetime running hours To convert Today running minutes into hours, simply divide by 60 $250/60 = 4.16$ Hrs.	Hrs.	23725.45	Must Have
TEMP	Inverter Temperature	Deg C	59.31	Must Have
FREQ	Frequency	Hz	50.02	Must Have
PF	Power Factor	-	0.93	Must Have
APOW	Apparent power	kVA	2.14	Must Have
RPOW	Reactive power	kVAr	0.23	Must Have
FT1	Fault-1 Normal: 0 VAC High: 1 VAC Low: 2 Grid Voltage Fail: 3 Voltage un-balance: 4	-	1	Must Have
FT2	Fault-2 Normal :0 Frequency High: 1 Frequency Low: 2	-	2	Must Have

FT3	Fault-3 Normal :0 AC Output overload :1 PV Short Circuit :2 AC Short circuit fault: 3 Leakage Current High: 4	-	0	Must Have
FT4	Fault-4 Normal :0 High Temperature: 1	-	0	Must Have
FT5	Fault-5 Normal: 0 Other Fault: 1	-	1	Must Have

11.3 Annexure-3 Single Phase-Hybrid Inverter (IH)

Annexure-3 Single Phase-Hybrid Inverter (IH)					
Message Name : Periodic Push Hybrid Inverter Data					
Message Format : JSON					
Message Type : Data					
Message Command Flow : Not Applicable for Data periodic Push					
Message response Flow : RMS -> Centralised IoT Platform					
Message Medium : Cellular					
Keyword structure :“{Part-1: Layer Identifier}- {Part-2: Parameter Identifier}”					
Combined Sample Keyword : IH-10-2-3- -POW					
Sample Keyword Part-2: Parameter Identifier	Sample Description	Samp le Unit	Sample Value	Sample Requirement in JSON	
VD	Virtual Device Index/Group	-	5	Must Have	
TIMESTAMP	RTC timestamp of RMS against all parameters of VD/group YYYY-MM-DD HH:MM: SS	-	2025-04-22 17:58:25	Must Have	
MAXINDEX	maximum index of local storage date	-	96	Must Have	
INDEX	reference of local storage	-	7	Must Have	
LOAD	Local storage retrieval command & status	-	0	Must Have	
STINTERVAL	Periodic interval at which RMS shall store and transmit data to server. (in minutes)	-	15	Must Have	
MSGID	Message Transaction Id - required for “Ondemand”/” Config” message type, request/response/acknowledgement/feedback	-	-	Must Have	
DATE	local storage date YYMMDD	-	250422	Must Have	
IMEI	IMEI No. of First Sim to be considered always for unique identity of RMS	-	123456123456123	Must Have	
ASN_33	Inverter Serial Number		-	33123450	Must Have
	RMS	0			
	DAQ	1-9			
	Meter (Optional)	21-29			
	Inverter, String Inverter /String Combiner Box (Optional)	31-50			
IST	Inverter Status OFF: 0, ON: 1, FAULT :2, Other: 3	-	1	Must Have	
DCV1	DC-1 Voltage (Similar for Up to N DC Channel Voltage)	Volts	331.20	Must Have	

DCI1	DC-1 Current (Similar for Up to N DC Channel Current)	Amp	3.04	Must Have
DCKW1	DC-1 Power (Similar for Up to N DC Channel Power)	kW	1.13	Must Have
VN	Phase to Neutral Voltage	Volts	255.80	Must Have
I	Phase current	Amp	3.91	Must Have
POW	Active Power	kW	0.99	Must Have
TKWH	Today Generated Energy	kWh	13.80	Must Have
TON	Today ON Time of Inverter To convert lifetime running minutes into hours, simply divide by 60 $250/60 = 4.16$ Hrs.	Hrs.	8.45	Must Have
LKWH	Lifetime Generated Energy	kWh	3019.90	Must Have
LON	Lifetime running hours To convert lifetime running minutes into hours, simply divide by 60 $250/60 = 4.16$ Hrs.	Hrs.	2075.9	Must Have
TEMP	Inverter Temperature	Deg C	52.31	Must Have
FREQ	Frequency	Hz	49.99	Must Have
PF	Power Factor	-	0.89	Must Have
APOW	Apparent power	kVA	1.04	Must Have
RPOW	Reactive power	kVAr	0.05	Must Have
BCHTKWH	Today Battery Charge Energy from PV	kWh	4.89	Good to Have
BCHLKWH	Lifetime Battery Charge Energy from PV	kWh	1212.34	Good to Have
IMODE	Inverter Mode 1- Power Feed in the Grid 2- Importing Power from Grid 3- No Power Imported from Grid 4- Islanding Mode	-	1	Good to Have
BST	Battery Status Healthy : 1 Faulty : 2	-	1	Good to Have
BCST	Charging Status Charging: 1 Discharging :2 Idle:3	-	2	Good to Have
BDCV	Battery Voltage	DC V	58.24	Good to Have
BDCI	Total Battery Current	DC I	15.12	Good to Have
BCH%	%Battery Charge	%	90.75	Good to Have
BHL%	% Battery Health	%	95.50	Good to Have
BTEMP	Battery Temperature	Deg C	55.12	Good to Have
BCV	Battery Charging Voltage	DC V	20.12	Good to Have
BCC	Battery Charging current	Amp	15.12	Good to Have

BWMOD	Working Mode 0: Self-consumption, 1: Peak shaving, 2: Battery Priority, 3: Load Shifting, 4: ToU Optimization, 5: Grid Support Mode, 6: PV Priority Mode, 7: Demand Response Mode	-	0	Good to Have
BDRT	Battery Discharging Rate	%	50.00	Good to Have
BDPWD	Battery Discharging Power Dispatching	%	15.50	Optional
BCHCTO%	Battery Charing Cut-off Capacity Range: 90 - 100 %	%	95.50	Good to Have
BDCHCTO%	Battery Discharging Cut-Off Capacity 12 -20%	%	15.50	Good to Have
FT1	Fault-1 Normal: 0 VAC High: 1 VAC Low: 2 Grid Voltage Fail: 3 Voltage un-balance: 4	-	1	Must Have
FT2	Fault-2 Normal :0 Frequency High: 1 Frequency Low: 2	-	2	Must Have
FT3	Fault-3 Normal :0 AC Output overload :1 PV Short Circuit :2 AC Short circuit fault: 3 Leakage Current High: 4	-	0	Must Have
FT4	Fault-4 Normal :0 High Temperature: 1	-	0	Must Have
FT5	Fault-5 Normal: 0 Other Fault: 1	-	1	Must Have

11.4 Annexure-4 Three Phase-Hybrid Inverter (IH)

Annexure-4 Three Phase-Hybrid Inverter (IH)					
Message Name : Periodic Push Hybrid Inverter Data					
Message Format : JSON					
Message Type : Data					
Message Command Flow : Not Applicable for Data periodic Push					
Message response Flow : RMS -> Centralised IoT Platform					
Message Medium : Cellular					
Keyword structure :“{Part-1: Layer Identifier}- {Part-2: Parameter Identifier}”					
Combined Sample Keyword : IH-10-2-3- -POW					
Sample Keyword Part-2: Parameter Identifier	Sample Description	Sample Unit	Sample Value	Sample Requirement in JSON	
VD	Virtual Device Index/Group	-	5	Must Have	
TIMESTAMP	RTC timestamp of RMS against all parameters of VD/group YYYY-MM-DD HH:MM: SS	-	2025-04-22 17:58:25	Must Have	
MAXINDEX	maximum index of local storage date	-	96	Must Have	
INDEX	reference of local storage	-	7	Must Have	
LOAD	Local storage retrieval command & status	-	0	Must Have	
STINTERVAL	Periodic interval at which RMS shall store and transmit data to server. (in minutes)	-	15	Must Have	
MSGID	Message Transaction Id - required for “Ondemand”/” Config” message type, request/response/acknowledgement/feedback	-	-	Must Have	
DATE	local storage date YYMMDD	-	250422	Must Have	
IMEI	IMEI No. of First Sim to be considered always for unique identity of RMS	-	123456123456123	Must Have	
ASN_34	Inverter Serial Number		-	34123450	Must Have
	RMS	0			
	DAQ	1-9			
	Meter (Optional)	21-29			
	Inverter, String Inverter /String Combiner Box (Optional)	31-50			
IST	Inverter Status OFF: 0, ON: 1, FAULT :2, Other: 3	-	1	Must Have	
DCV1	DC-1 Voltage (Similar for Up to N DC Channel Voltage)	Volts	372.80	Must Have	

DCI1	DC-1 Current (Similar for Up to N DC Channel Current)	Amp	3.92	Must Have
DCKW1	DC-1 Power (Similar for Up to N DC Channel Power)	kW	1.36	Must Have
RPHV	R phase voltage	Volts	253.20	Must Have
RPHI	R phase current	Amp	1.98	Must Have
POWR	R phase Active Power	kW	0.51	Must Have
YPHV	Y phase voltage	Volts	251.30	Must Have
YPHI	Y phase current	Amp	1.81	Must Have
POWY	Y phase Active Power	kW	0.49	Must Have
BPHV	B phase voltage	Volts	252.80	Must Have
BPHI	B phase current	Amp	1.81	Must Have
POWB	B phase Active Power	kW	0.45	Must Have
POW	Total Active Power	kW	1.13	Must Have
TKWH	Today Generated Energy	kWh	17.91	Must Have
TON	Today ON Time of Inverter To convert Today running minutes into hours, simply divide by 60 $250/60 = 4.16$ Hrs.	Hrs.	8.54	Must Have
LKWH	Lifetime Generated Energy	kWh	50629.34	Must Have
LON	Lifetime running hours To convert Today running minutes into hours, simply divide by 60 $250/60 = 4.16$ Hrs.	Hrs.	23725.87	Must Have
TEMP	Inverter Temperature	Deg C	59.31	Must Have
FREQ	Frequency	Hz	49.99	Must Have
PF	Power Factor	-	0.93	Must Have
APOW	Apparent power	kVA	1.16	Must Have
RPOW	Reactive power	kVAr	0.03	Must Have
BCHTKWH	Today Battery Charge Energy from PV	kWh	6.87	Good to Have
BCHLKWH	Lifetime Battery Charge Energy from PPV	kWh	4673.56	Good to Have
IMODE	Inverter Mode 1- Power Feed in the Grid 2- Importing Power from Grid 3- No Power Imported from Grid 4- Islanding Mode	-	1	Good to Have
BST	Battery Status Healthy: 1 Faulty: 2	-	1	Good to Have

BCST	Charging Status Charging: 1 Discharging :2 Idle:3	-	2	Good to Have
BDCV	Battery Voltage	DC V	58.24	Good to Have
BDCI	Total Battery Current	DC I	15.12	Good to Have
BCH%	%Battery Charge	%	90.75	Good to Have
BHL%	% Battery Health	%	95.50	Good to Have
BTEMP	Battery Temperature	Deg C	55.12	Good to Have
BCV	Battery Charging Voltage	DC V	20.12	Good to Have
BCC	Battery Charging current	Amp	15.12	Good to Have
BWMOD	Working Mode 0: Self-consumption, 1: Peak shaving, 2: Battery Priority, 3: Load Shifting, 4: ToU Optimization, 5: Grid Support Mode, 6: PV Priority Mode, 7: Demand Response Mode	-	0	Good to Have
BDRT	Battery Discharging Rate	%	50.00	Good to Have
BDPWD	Battery Discharging Power Dispatching	%	15.50	Optional
BCHCTO%	Battery Charing Cut-off Capacity Range: 90 - 100 %	%	95.50	Good to Have
BDCHCTO %	Battery Discharging Cut-Off Capacity 12 -20%	%	15.50	Good to Have
FT1	Fault-1 Normal: 0 VAC High: 1 VAC Low: 2 Grid Voltage Fail: 3 Voltage un-balance: 4	-	1	Must Have
FT2	Fault-2 Normal :0 Frequency High: 1 Frequency Low: 2	-	2	Must Have
FT3	Fault-3 Normal :0 AC Output overload :1 PV Short Circuit :2 AC Short circuit fault: 3 Leakage Current High: 4	-	0	Must Have
FT4	Fault-4 Normal :0 High Temperature: 1	-	0	Must Have
FT5	Fault-5 Normal: 0 Other Fault: 1	-	1	Must Have

11.5 Annexure-5: Single Phase Energy Meter (Optional)

Annexure-5: Single Phase Energy Meter (Optional)

Message Name

: Periodic Push Energy Meter Data

Message Format

: JSON

Message Type

: Data

Message Command Flow

: Not Applicable for Data periodic Push

Message response Flow

: RMS -> Centralised IoT Platform

Message Medium

: Cellular

Keyword structure

: "{Part-1: Layer Identifier}- {Part-2: Parameter Identifier}"

Combined Sample Keyword

: MS-10-2-3- -POW (For Solar Meter)

: MC-10-2-3- -POW (For Consumption Meter)

: MN-10-2-3- -POW (For Net Meter)

Sample Keyword Part-2: Parameter Identifier	Sample Description	Sample Unit	Sample Value	Sample Requirement in JSON								
VD	Virtual Device Index/Group	-	5	Must Have								
TIMESTAMP	RTC timestamp of RMS against all parameters of VD/group YYYY-MM-DD HH:MM: SS	-	2025-12-02 12:58:25	Must Have								
MAXINDEX	maximum index of local storage date	-	96	Must Have								
INDEX	reference of local storage	-	7	Must Have								
LOAD	Local storage retrieval command & status	-	0	Must Have								
STINTERVAL	Periodic interval at which RMS shall store and transmit data to server. (in minutes)	-	15	Must Have								
MSGID	Message Transaction Id - required for "Ondemand"/" Config" message type, request/response/acknowledgement/ Feedback	-	-	Must Have								
DATE	local storage date YYMMDD	-	251202	Must Have								
IMEI	IMEI No. of First Sim to be considered always for unique identity of RMS	-	12345612345 6123	Must Have								
ASN_21	Meter Serial Number <table> <tr> <td>RMS</td> <td>0</td> </tr> <tr> <td>DAQ</td> <td>1-9</td> </tr> <tr> <td>Meter</td> <td>21-29</td> </tr> <tr> <td>Inverter, String Inverter /String Combiner Box</td> <td>31-50</td> </tr> </table>	RMS	0	DAQ	1-9	Meter	21-29	Inverter, String Inverter /String Combiner Box	31-50	-	21123450	Must Have
RMS	0											
DAQ	1-9											
Meter	21-29											
Inverter, String Inverter /String Combiner Box	31-50											

MTDET	Meter Detail		30012302	Must Have
MTBLDATE	Billing Date for meter 1		15	Must Have
DATE	Present date for meter1		251202	Must Have
TIME	Present time for meter1		125400	Must Have
VN	Phase to Neutral Voltage	Volts	244.66	Must Have
I	Phase Current	Amp	0.94	Must Have
POW	Active Power	kW	0.23	Must Have
RPOW	Reactive Power	kVAr	0.02	Must Have
APOW	Apparent Power	kVA	0.25	Good to Have
PF	Average PF	-	0.98	Must Have
FRQ	Grid Frequency	Hz	49.89	Must Have
KWHNET	Cumulative Net Energy	kWh	5695.92	Good to Have
KWHIMP	Cumulative Import Energy	kWh	6046.70	Must Have
KWHEXP	Cumulative Export Energy	kWh	347.78	Must Have
KVAHNET	Cumulative Net Energy	kVAh	6194.35	Good to Have
KVAHIMP	Cumulative Import Energy	kVAh	6606.80	Good to Have
KVAHEXP	Cumulative Export Energy	kVAh	412.45	Good to Have
MDKWIMP	Rising Demand (Import)	kW	2.12	Must Have
MDKWEXP	Rising Demand (Export)	kW	0.41	Must Have
POFF	Grid Power Failure in Minutes	Min	120	Must Have
TC	Total Tamper Counts	Nos	117	Good to Have
LBKWHNET	Last Billing Cycle Net Energy	kWh	5609.33	Optional
LBKWHIMP	Last Billing Cycle Import Energy	kWh	5934.45	Optional
LBKWHEXP	Last Billing Cycle Export Energy	kWh	325.12	Optional
PMDKVAIMP	Present MD KVA Import	kVA	1.95	Optional
PMDKVAEXP	Present MD KVA Export	kVA	0	Optional
LBMDKWIMP	Last Billing MD KW Import	kW	2.09	Optional
LBMDKWEXP	Last Billing MD KW Export	kW	0	Optional
LBMDKVAIMP	Last Billing MD KVA Import	kVA	2.18	Optional
LBMDKVAEXP	Last Billing MD KVA Export	kVA	0	Optional
MDRSTC	MD Reset Count	Nos	36	Optional

11.6 Annexure-6: Three Phase Energy Meter

Annexure-6: Three Phase Energy Meter (Optional)				
Message Name : Periodic Push Energy Meter Data Message Format : JSON Message Type : Data Message Command Flow : Not Applicable for Data periodic Push Message response Flow : RMS -> Centralised IoT Platform Message Medium : Cellular Keyword structure : "{Part-1: Layer Identifier}- {Part-2: Parameter Identifier}" Combined Sample Keyword : MS-10-2-3- -POW (For Solar Meter) : MC-10-2-3- -POW (For Consumption Meter) : MN-10-2-3- -POW (For Net Meter)				
Sample Keyword Part-2: Parameter Identifier	Sample Description	Sample Unit	Sample Value	Sample Requirement in JSON
VD	Virtual Device Index/Group	-	5	Must Have
TIMESTAMP	RTC timestamp of RMS against all parameters of VD/group YYYY-MM-DD HH:MM: SS	-	2025-12-02 12:58:25	Must Have
MAXINDEX	maximum index of local storage date	-	96	Must Have
INDEX	reference of local storage	-	7	Must Have
LOAD	Local storage retrieval command & status	-	0	Must Have
STINTERVAL	Periodic interval at which RMS shall store and transmit data to server. (in minutes)	-	15	Must Have
MSGID	Message Transaction Id - required for "Ondemand"/" Config" message type, request/response/acknowledgement / Feedback	-	-	Must Have
DATE	local storage date YYMMDD	-	251202	Must Have
IMEI	IMEI No. of First Sim to be considered always for unique identity of RMS	-	123456123456 123	Must Have

ASN_22	Meter Serial Number		-	33123450	Must Have
	RMS	0			
	DAQ	1-9			
	Meter	21-29			
	Inverter, String Inverter /String Combiner Box	31-50			
MTDET	Meter Detail			30012302	Must Have
MTBLDATE	Billing Date for meter 1			15	Must Have
DATE	Present date for meter1		YYMMDD	251202	Must Have
TIME	Present time for meter1		HHMMSS	125400	Must Have
IR	R Phase Current		Amp	0.94	Must Have
IY	Y Phase Current		Amp	1.01	Must Have
IB	B Phase Current		Amp	1.04	Must Have
VRN	R Phase to Neutral Voltage		Volts	241.26	Must Have
VYN	Y Phase to Neutral Voltage		Volts	240.78	Must Have
VBN	B Phase to Neutral Voltage		Volts	239.79	Must Have
VRV	Phase to Phase Voltage(R-Y)		Volts	415.67	Must Have
VYB	Phase to Phase Voltage(Y-B)		Volts	419.23	Must Have
VBR	Phase to Phase Voltage(B-R)		Volts	417.89	Must Have
PFR	R Phase Power Factor		-	0.99	Must Have
PFY	Y Phase Power Factor		-	0.99	Must Have
PFB	B Phase Power Factor		-	0.99	Must Have
FRQ	Grid Frequency		Hz	50.01	Must Have
POWR	R Phase Active Power		kW	0.22	Must Have
POWY	Y Phase Active Power		kW	0.24	Must Have
POWB	B Phase Active Power		kW	0.24	Must Have
POW	Total Active Power		kW	0.74	Must Have
RPOWR	R Phase Reactive Power		kVAr	0.02	Good to Have
RPOWY	Y Phase Reactive Power		kVAr	0.04	Good to Have
RPOWB	B Phase Reactive Power		kVAr	0.03	Good to Have
RPOW	Total Reactive Power		kVAr	0.09	Must Have
APOWR	R Phase Apparent Power		kVA	0.25	Good to Have
APOWY	Y Phase Apparent Power		kVA	0.24	Good to Have
APOWB	B Phase Apparent Power		kVA	0.27	Good to Have
APOW	Total Apparent Power		kVA	0.79	Must Have
KWHNET	Cumulative Net Energy		kWh	10063.49	Good to Have
KWHIMP	Cumulative Import Energy		kWh	13015.89	Must Have
KWHEXP	Cumulative Export Energy		kWh	2952.4	Must Have
KVAHNET	Cumulative Net Energy		kVAh	12027.35	Must Have
KVAHIMP	Cumulative Import Energy		kWh	15239.78	Must Have
KVAHEXP	Cumulative Export Energy		kWh	3212.43	Must Have
MDKWIMP	Rising Demand (Import)		kW	12.45	Must Have

MDKWEXP	Rising Demand (Export)	kW	3.43	Must Have
POFF	Grid Power Failure in Minutes	Mins	120	Must Have
TC	Total Tamper Counts	Nos	143	Good to Have
PF	Average PF	-	0.98	Must Have
LBKWHNET	Last Billing Cycle Net Energy	kWh	10060.49	Optional
LBKWHIMP	Last Billing Cycle Import Energy	kWh	13010.89	Optional
LBKWHEXP	Last Billing Cycle Export Energy	kWh	2950.4	Optional
PMDKVAIMP	Present MD KVA Import	kVA	11.97	Optional
PMDKVAEXP	Present MD KVA Export	kVA	3.45	Optional
LBMDKWIMP	Last Billing MD KW Import	kVA	13.44	Optional
LBMDKWEXP	Last Billing MD KW Export	kVA	4.34	Optional
LBMDKVAIMP	Last Billing MD KVA Import	kVA	14.32	Optional
LBMDKVAEXP	Last Billing MD KVA Export	kVA	5.34	Optional
MDRSTC	MD Reset Count	Nos	14	Optional

11.7 Annexure-7 String Combiner Box

Annexure-7 String Combiner Box

Annexure-7 String Combiner Box (Optional)

Message Name

: Periodic Push String Combiner Box

Message Format

: JSON

Message Type

: Data

Message Command Flow

: Not Applicable for Data periodic Push

Message response Flow

: RMS -> Centralised IoT Platform

Message Medium

: Cellular

Keyword structure

: “{Part-1: Layer Identifier}- {Part-2: Parameter Identifier}”

Combined Sample Keyword

: S-99-21-1-1-I1

Sample Keyword Part-2: Parameter	Sample Description	Sample Unit	Sample Value	Sample Requirement in JSON								
VD	Virtual Device Index/Group	-	9	Must Have								
TIMESTAMP	RTC timestamp of RMS against all parameters of VD/group YYYY-MM-DD HH:MM: SS	-	2025-07-07 02:08:55	Must Have								
MAXINDEX	maximum index of local storage date	-	96	Must Have								
INDEX	reference of local storage	-	7	Must Have								
LOAD	Local storage retrieval command & status	-	0	Must Have								
STINTERVAL	Periodic interval at which RMS shall store and transmit data to server. (in minutes)	-	15	Must Have								
MSGID	Message Transaction Id - required for “Ondemand”/” Config” message type, request/response/acknowledgement/ feedback	-	-	Must Have								
DATE	local storage date YYMMDD	-	250707	Must Have								
IMEI	IMEI No. of First Sim to be considered always for unique identity of RMS	-	1234561234 56123	Must Have								
ASN_41	DAQ Serial Number <table> <tr> <td>RMS</td> <td>0</td> </tr> <tr> <td>DAQ</td> <td>1-9</td> </tr> <tr> <td>Meter</td> <td>21-29</td> </tr> <tr> <td>Inverter, String Inverter /String Combiner Box</td> <td>31-50</td> </tr> </table>	RMS	0	DAQ	1-9	Meter	21-29	Inverter, String Inverter /String Combiner Box	31-50	-	34123450	Must Have
RMS	0											
DAQ	1-9											
Meter	21-29											
Inverter, String Inverter /String Combiner Box	31-50											

I1	SCB1, Channel1 Current	Amp	30.11	Must Have
I2	SCB1, Channel2 Current	Amp	50.11	Must Have
I3	SCB1, Channel3 Current	Amp	40.11	Must Have
I4	SCB1, Channel4 Current	Amp	50.11	Must Have
I5	SCB1, Channel5 Current	Amp	50.11	Must Have
I6	SCB1, Channel6 Current	Amp	5.11	Must Have
I7	SCB1, Channel7 Current	Amp	5.11	Must Have
I8	SCB1, Channel8 Current	Amp	5.11	Must Have
IN	SCB1, Channel N Current (Up to N Channel)	Amp	5.11	Must Have
DCV	SCB1, DC Voltage	Volts	635.02	Must Have
DCTOTI	SCB1, Total DC Current	Amp	40.02	Must Have
DCTOTKW	SCB1, Total DC Power	kW	28.01	Must Have
DI1	SCB1, Digital Input1	-	1	Must Have
DI2	SCB1, Digital Input2	-	1	Must Have
T1	SCB1, Temperature1	Deg C	35.00	Must Have
T2	SCB1, Temperature2	Deg C	37.50	Must Have
T3	SCB1, Temperature3	Deg C	34.65	Must Have

11.8 Annexure-8 DAQ System

Annexure-8 DAQ System

Message Name

: Periodic Push DAQ

Message Format

: JSON

Message Type

: Data

Message Command Flow

: Not Applicable for Data periodic Push

Message response Flow

: RMS -> Centralised IoT Platform

Message Medium

: Cellular

Keyword structure

: "{Part-1: Layer Identifier}- {Part-2: Parameter Identifier}"

Combined Sample Keyword

: D-1-0- - -FDRST1

Sample Keyword Part-2: Parameter Identifier	Sample Description	Sample Unit	Sample Value	Sample Requirement in JSON	
VD	Virtual Device Index/Group	-	5	Must Have	
TIMESTAMP	RTC timestamp of RMS against all parameters of VD/group YYYY-MM-DD HH:MM: SS	-	2025-04-22 17:58:25	Must Have	
MAXINDEX	maximum index of local storage date	-	96	Must Have	
INDEX	reference of local storage	-	7	Must Have	
LOAD	Local storage retrieval command & status	-	0	Must Have	
STINTERVAL	Periodic interval at which RMS shall store and transmit data to server. (in minutes)	-	15	Must Have	
MSGID	Message Transaction Id - required for "Ondemand"/"Config" message type, request/response/acknowledgement/ Feedback	-	-	Must Have	
DATE	local storage date (YYMMDD)	-	250422	Must Have	
IMEI	IMEI No. of First Sim to be considered always for unique identity of RMS	-	123456412345613	Must Have	
ASN_1	DAQ Serial Number		-	1123450	Must Have
	RMS	0			
	DAQ	1-9			
	Meter	21-29			
	Inverter, String Inverter /String Combiner Box	31-50			

MODTMP1	Module Temperature-1 (0-150)	Deg C	60.12	Must Have
MODTMP2	Module Temperature-2 (0-150)	Deg C	61.32	Good to Have
HUMDT	Humidity (0-100)	%	65.32	Good to Have
AMBTMP1	Ambient Temperature-1 (0-150)	Deg C	45.54	Must Have
AMBTMP2	Ambient Temperature-2 (0-150)	Deg C	46.64	Good to Have
RAING	Rain Gauge (0-50)	mm	23.34	Good to Have
POA1	Plan of Array Irradiation-1(0-2000)	w/m2	987.23	Must Have
POA2	Plan of Array Irradiation-2 (0-2000)	w/m2	902.34	Good to Have
GHI1	Global Horizontal Irradiation-1 (0-2000)	w/m2	876.45	Must Have
GHI2	Global Horizontal Irradiation-2 (0-2000)	w/m2	845.65	Good to Have
WDIR	Wind Direction (0-360)	Degree	120.43	Good to Have
WSPEED	Wind Speed (0-60)	m/s	23.45	Good to Have
YLDPOA1	Plan of Array 1 Radiation Yield (0-9999999)	kWh/m2	3423.34	Good to Have
YLDPOA2	Plan of Array 2 Radiation Yield (0-9999999)	kWh/m2	3245.23	Good to Have
YLDGHI1	Global Horizontal2 Radiation Yield (0-9999999)	kWh/m2	2789.34	Good to Have
YLDGHI2	Global Horizontal2 Radiation Yield (0-9999999)	kWh/m2	2867.23	Good to Have
DHRPOA1	POA-1 Day First Run Minutes To convert running seconds into Mins, simply divide by 60 $250/60 = 4.16$ Min.	Min	56.78	Must Have
THRPOA1	POA-1 Lifetime Run Minutes	Min	6537.45	Must Have
DHRPOA2	POA-2 Day First Run Minutes	Min	56.78	Good to Have
THRPOA2	POA-2 Lifetime Run Minutes	Min	6537.45	Good to Have
DHRGHI1	GHI-1 Day First Run Minutes	Min	45.56	Must Have
THRGHI1	GHI-1 Lifetime Run Minutes	Min	3545.45	Must Have
DHRGHI2	GHI-2 Day First Run Minutes	Min	45.56	Good to Have
THRGHI2	GHI-2 Lifetime Run Minutes	Min	3545.45	Good to Have
MCCBST1	DB MCCB1 Status: OFF-0, ON-1 (Similar for Up to N MCCB Status)	-	1	Good to Have
FDRST1	Feeder1 Status: OFF-0, ON-1 (Similar for Up to N Feeder Status)	-	1	Good to Have
TRPST1	Transformer1 HV Side Feeder Status: OFF-0, ON-1 (Similar for Up to N TR HV Status)	-	1	Good to Have
TRSST1	Transformer1 LV Side Feeder Status: OFF-0, ON-1 (Similar for Up to N TR LV Status)	-	1	Good to Have
LNST1	Line1 Feeder Status: OFF-0, ON-1 (Similar for Up to N Line Status)	-	1	Good to Have

11.9 Annexure-9 RMS

Annexure-9 RMS				
Message Name : RMS Message Format : JSON Message Type : Heartbeat Message Command Flow : Not Applicable Message response Flow : RMS -> Centralised IoT Platform Message Medium : Cellular Keyword structure : "{Part-1: Layer Identifier}- {Part-2: Parameter Identifier}" Combined Sample Keyword : R-1-0- - -RSSI				
Sample Keyword Part-2: Parameter	Sample Description	Sample Unit	Sample Value	Sample Requirement in JSON
VD	Virtual Device Index/Group	-	0	Must Have
TIMESTAMP	RTC timestamp of RMS against all parameters of VD/group	-	2025-04-22 17:58:25	Must Have
MAXINDEX	maximum index of local storage date	-	96	Must Have
INDEX	reference of local storage	-	7	Must Have
LOAD	Local storage retrieval command & status	-	0	Must Have
STINTERVAL	Periodic interval at which RMS shall store and transmit data to server. (in minutes)	-	15	Must Have
MSGID	Message Transaction Id - required for "Ondemand"/" Config" message type, request/response/acknowledgement/feedback	-	-	Must Have
DATE	local storage date YYMMDD	-	250422	Must Have
IMEI	IMEI No. of First Sim to be considered always for unique identity of RMS	-	123456123 456123	Must Have
ASN_0	RMS Serial Number		10123450	Must Have
	RMS	0		
	DAQ	1-9		
	Meter	21-29		
	Inverter, String Inverter /String Combiner Box	31-50		
GSM	Device connected to GSM network	-	1	Must Have
SIM	SIM detected (1 - detected)	-	1	Must Have
NET	Device in Network (1 - in network)	-	1	Must Have
GPRS	GPRS connected (1 - connected)	-	1	Must Have

RSSI	Signal Strength	-	22	Must Have
SD	SD card detected (1 – detected,0-Not detected)	-	1	Good to Have
ONLINE	Device Online (1- Online, 0-Offline)	-	1	Good to Have
RF	RF Module Status (1-ON,0-OFF)	-	1	Must Have
RTCDATE	RTC Date (YYMMDD)	-	250422	Must Have
RTCTIME	RTC Time (HHMMSS)	-	175855	Must Have
TEMP	RMS Device Temperature	Deg C	45.5	Must Have
LAT	Latitude (XX. XXXXXX)	-	19.066645	Good to Have
LONG	Longitude (XX. XXXXXX)	-	72.877865	Good to Have
SIMSLOT	Sim Slot (Current Sim Slot: 1 or 2)	-	1	Must Have
SIMCHNGCN	Total Sim Slot Change Count	-	10	Good to Have
FLASH	Device Flash Status 1: Detected 0: Error	-	1	Good to Have
BATTST	Battery Input Status: 1 if on battery power else 0	-	0	Optional
VBATT	Battery Voltage	Volts	5.5	Optional
PST	Power Supply (1-Mains, 2-Battery)	-	1	Optional

11.10 Annexure-10 Command & Control

Annexure-10 Command & Control				
Message Name : On Demand Read/Write Parameter Message Format : JSON Message Type : Config/Ondemand Message Command Flow : Cloud Server -> RMS Message response Flow : RMS -> Cloud Server Message Medium : Cellular Keyword structure : "{Keyword Part-1 : Identifier}- {Keyword Part-2 Parameter}" Combined Sample Keyword : IG-10-2-3- -POWSP				
Sample Keyword Part-2: Parameter	Sample Description	Sample Unit	Sample Value	Sample Requirement in JSON
TIMESTAMP	RTC timestamp of RMS against all parameters of VD/group	-	2025-04-22 17:58:25	Must Have
TYPE	config/ondemand	-	config	Must Have
CMD	Read/Write	-	read	Must Have
MSGID	Message Transaction Id - required for "Ondemand"/" Config" message type, request/response/acknowledgement/feedback	-	"1234"	Must Have
VSP	Inverter Voltage Set Point	V	265.50	Good to Have For configuration
POWSP	Inverter Active Power Set Point	KW	25.50	Good to Have For configuration
POWSPPER	Inverter Active Power Limit Ratio (%)	%	100.00	Good to Have For configuration
RPOWSP	Inverter Reactive Power Set Point	KVAr	1.5	Good to Have For configuration
PFSP	Inverter Power Factor Set Point	-	0.98	Good to Have For configuration

INVCMD	Inverter Control 1 – ON, 2 – OFF, 3 – Emergency Shutdown	-	1	Good to Have For control
CBCHCTO	Charing Cut-off Capacity Range: 90 - 100 %	%	95.00	Good to Have For configuration
CBDCHCTO	Discharging Cut-Off Capacity Range: 12-20%	%	15.00	Good to Have For configuration
CIMODE	Hybrid Inverter Mode Control 1- Power Feed in the Grid 2- Importing Power from Grid 3- No Power Imported from Grid 4- Islanding Mode	-	1	Good to Have For control
CBCHDCH	Battery Charge/Discharge Command 1: Charge, 2: Discharge, 3: Stop	-	1	Good to Have For control
FDRONCMD1	Feeder1 On Command (Similar for Up to N Feeder)	-	1	Good to Have For control
FDROFFCMD1	Feeder1 OFF Command (Similar for Up to N Feeder)	-	1	Good to Have For control
TRPONCMD1	Transformer 1 Primary on Command (Similar for Up to N Transformer)	-	1	Good to Have For control
TRPOFFCMD1	Transformer 1 Primary Off Command (Similar for Up to N Transformer)	-	1	Good to Have For control
TRSONCMD1	Transformer 1 Secondary on Command (Similar for Up to N Transformer)	-	1	Good to Have For control
TRSOFFCMD1	Transformer 1 Secondary OFF Command (Similar for Up to N Transformer)	-	1	Good to Have For control
APN1	APN Value for sim1	-	m2misafe	Must Have For configuration
USR1	sim1 user name	-		Optional
PASS1	sim1 password	-		Optional
APN2	APN Value for sim2	-	m2misafe	Must Have For configuration

USR2	Sim2 user name	-		Optional
PASS2	Sim2 password	-		Optional
RESTART	To restart RMS, 1: Execute command	-		Must Have For control
UPDATE INTERVAL	Enter update interval in minutes.	-		Must Have For configuration
HEART INTERVAL	Heartbeat Update Interval in mins	-		Must Have For configuration
URTCDATE	RMS RTC Date (YYMMDD) Update RMS RTC Time (HH:MM: SS) Update	-		Must Have For configuration
URTCTIME	RMS RTC Time (HH:MM: SS) Update - 24-hour format	-		Must Have For configuration
UPDATERTC	Update RTC, 1: Execute command, 0 : Successful execution	-		Must Have For configuration
GSMSYNC	RTC auto GSM synchronization, 1: to execute command	-		Must Have For configuration
AI1ZERO	Engineering Zero Value (4 mA dc) for AI1 E.G. 0(LPM)	-		Must Have For configuration
AI1SPAN	Engineering Span Value (20 mA dc) for AI1 E.G. 5000(LPM)	-		Must Have For configuration
URL	URL of Device Management Server	-		Must Have For configuration
PORT	Port of Device Management Server	-		Must Have For configuration
CID	Unique Client id of device	-		Must Have For configuration
USERNAME	Username for device authentication	-		Must Have For configuration
PASSWORD	Password for device authentication	-		Must Have For configuration
FTPURL	URL for FTP	-		Must Have For configuration
FTPUSER	Username for FTP	-		Must Have

				For configuration
FTPPASS	Password for FTP	-		Must Have For configuration
FTPSPORT	Port for FTP	-		Must Have For configuration
FTPDOWN	Download Certificates from ftp 1: To execute command, 0: Command is successfully executed	-		Must Have For configuration

11.11 Annexure-11: Sample JSON Format of Annexure-1 (Single Phase-String Inverter (IS))

Annexure-11: Sample JSON Format of Annexure-1 (Single Phase-String Inverter (IS))	
<pre>{ "VD":5, "TIMESTAMP":"2025-07-07 17:58:29", "MAXINDEX": 72, "INDEX":72, "LOAD": 0, "STINTERVAL":15, "MSGID": "", "DATE": 250707, "IMEI": "123456789101112", "ASN_32":34123450, "IS-10-2-3--IST":1, "IS-10-2-3--DCV1":342.38, "IS-10-2-3--DCI1":6.01, "IS-10-2-3--DCKW1":2.7, "IS-10-2-3--RPHV":254.10, "IS-10-2-3--RPHI":2.77, "IS-10-2-3--POWR":0.70, "IS-10-2-3--YPHV":252.10, "IS-10-2-3--YPHI":2.77, "IS-10-2-3--POWY":0.71, "IS-10-2-3- -BPHV":253.90, "IS-10-2-3--BPHI":2.77, "IS-10-2-3--POWB":0.71, "IS-10-2-3--POW":1.95, "IS-10-2-3--TKWH":17.40, "IS-10-2-3--TON":8.16, "IS-10-2-3--LKWH":50628.45, "IS-10-2-3--LON":23725.45, "IS-10-2-3--TEMP":59.39, "IS-10-2-3--FREQ":50.02,</pre>	

```

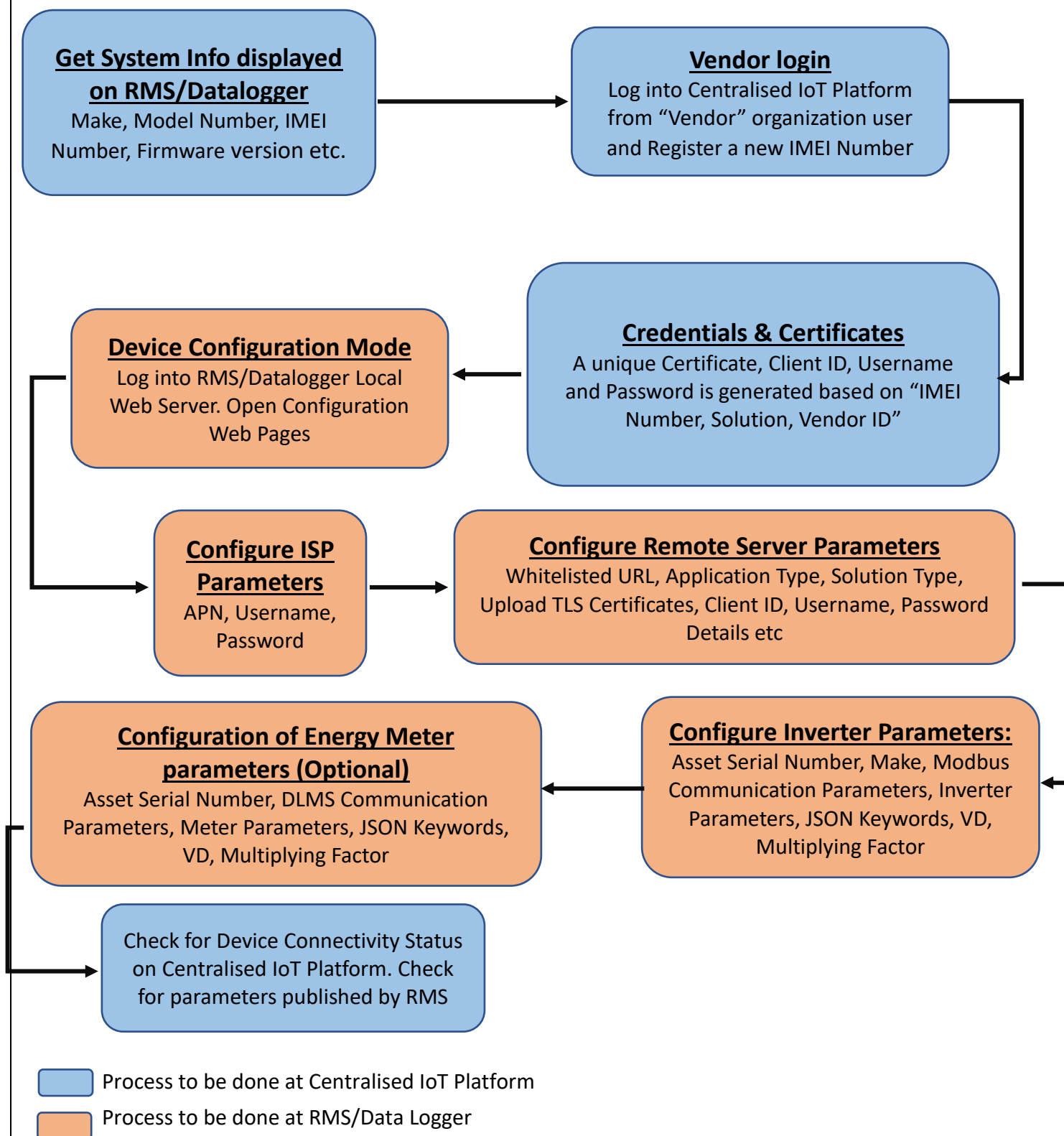
"IS-10-2-3--PF":0.93,
"IS-10-2-3--APOW":2.14
"IS-10-2-3--RPOW":0.23,
"IS-10-2-3--FT1":1,
"IS-10-2-3--FT2":2,
"IS-10-2-3--FT3":0,
"IS-10-2-3--FT4":0,
"IS-10-2-3--FT5":1
}

```

11.12 Annexure-12: Sample JSON Format of Annexure-10 (Command)

Annexure-12: Sample JSON Format of Annexure-10 (Command)
Command Message: Inverter “Voltage Limit” Command from Centralised IoT Platform to RMS <pre> { "msgid": "6372587", "CMKEY ":"963213", "PMKEY ":"380670", "timestamp": "2025-12-02 18:03:58", "type": "ondemand", "cmd": "write", "IS-1-0---VSP": 287 } </pre>
Response Message: RMS to Centralised IoT Platform <pre> { "msgid":"6372588", " CMKEY ":"963213", " PMKEY ":"380670", "timestamp":"2025-12-02 18:04:00", "type":"ondemand", "cmd":"read", "IS-1-0---VSP":287, "IS-1-0---POWSP":0, "IS-1-0---RPOWSP":0, "IS-1-0---PFSP":1, "IS-1-0---POWSPPER":100 } </pre>

11.13 Annexure-13: SOP for Registration and Issuance of certificate for RMS/Datalogger



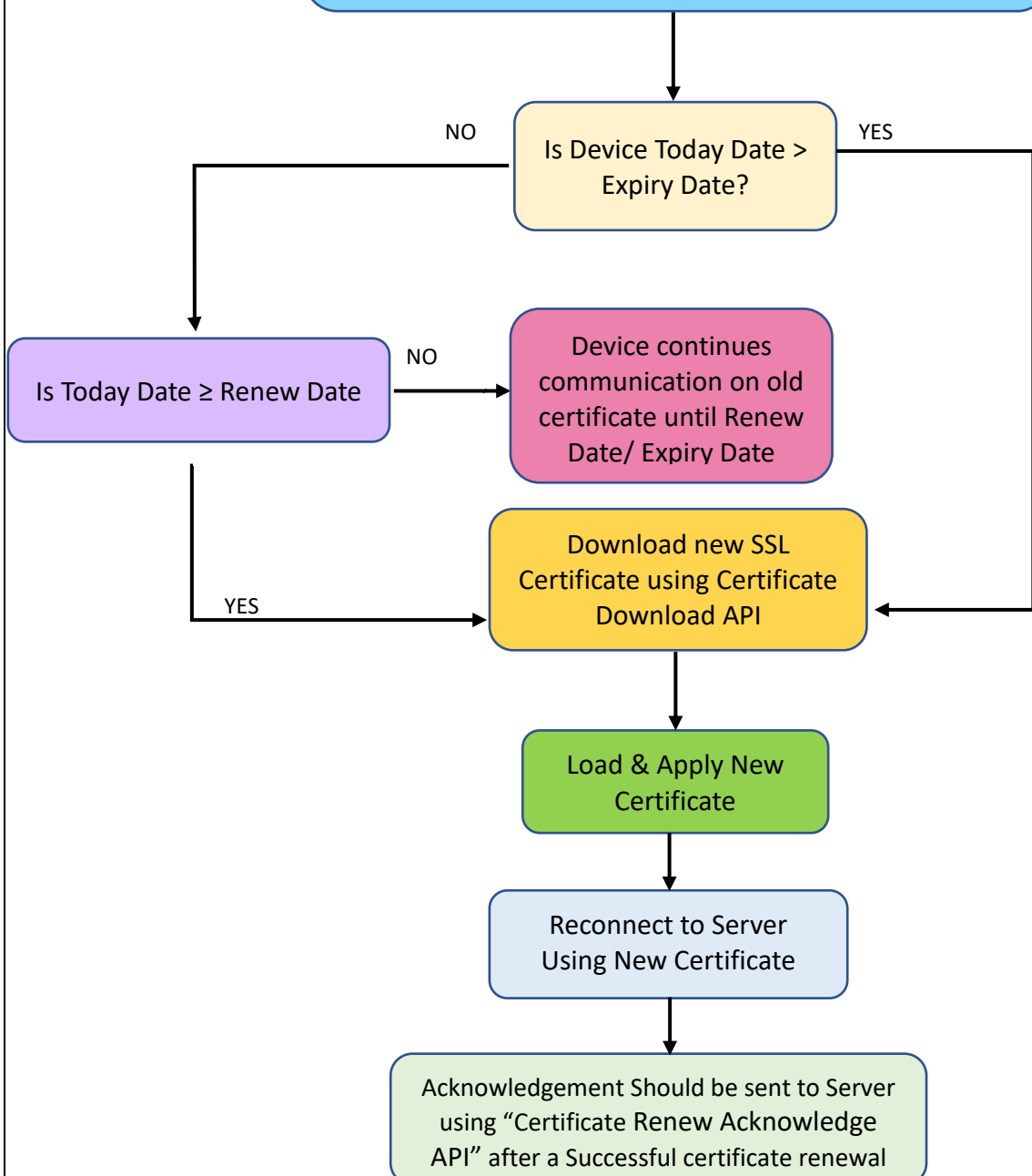
11.14 Annexure-14: SOP for update of certificate

Device Subscribes to Info Topic

rtsg-1/ongridrooftop/IMEI/info/sub

Device Receives:

IMEI Number, Username & Password, HTTPS URL, HTTPS Port, HTTPs Username, HTTPS Password, Certificate Download API, SSL Certificate Expiry Date, SSL Certificate Renew Date, Certificate Renew Acknowledge API



Note: There is a **1-month gap** between the Renew Date and Expiry Date.

If the device **does not update the new certificate before the Expiry Date**, the old certificate becomes invalid and the **device will lose server connection**.